Designing a Gamified Application for User Privacy Preference Extraction

Continual Assessment Report

Lee Chen Xuan

A0276149U

School of Computing

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Abstract

[Write this after the main sections are drafted.]

Contents

1 Introduction **2**

1.1 Background 2

1.2 Project Objectives 2

1.3 Ideation of Solution 3

1.3.1 Requirements 3

1.3.2 Constructing the Solution 3

1.3.3 Constraints 3

2 Literature Review **4**

2.1 Review approach 4

2.2 Key themes 4

2.2.1 [Theme 1] 4

2.2.2 [Theme 2] 4

2.3 Research gap and implications for this project 4

3 Progress to Date **5**

3.1 Work completed 5

3.2 Design or investigation alternatives 5

3.3 Preliminary design or investigation 5

3.4 Challenges and risks 5

4 Research Plan for Next Semester **6**

4.1 Planned work and methods 6

4.2 Milestones 6

4.3 Evaluation and success criteria 6

5 Conclusion **7**

Chapter 1

Introduction

1.1 Background

Even in this day and age of absurd technological advancements, user privacy preference remains an alien topic for the general public. There are several ways user privacy preferences can be collected online, some common ways that are used may be through surveys [1], device settings parsing [2], and cookies' consent banner interaction [3]. This project will specifically be exploring how user privacy preferences are extracted through cookies and their consent banners, commonly presented to users with the interface of a popup provided by CMPs (Consent Management Platforms) called cookie banners [4].

Most people don't really pay close attention to the user privacy preferences that they (have to) set each time they visit a new website on the Internet and is presented with a cookie banner. Whenever people are presented with a popup-like interface with verbose texts that they have to read through, the instinct for most people would be to get rid of the popup window. As such when combined with the assumption that the general public isn't sufficiently educated on the topic of the importance of user privacy preferences and how to manage them well, users tend to click the options presented on the interface that allows them to proceed . Some reasons why people tend to ignore how user privacy preference extraction works in cookie banners is likely driven by the cookie banners' complex design, knowledge limitations, and conveniency [5].

In view of this problem presented and potential causes that are established, this project aims to solve this issue by gamifying the process of user privacy preference extraction.

1.2 Project Objectives

The aim of this project is to [one-sentence overall aim]. The objectives are:

action or outcome.

Objective 2.

Objective 3.

1.3 Ideation of Solution

[State what is in scope, important constraints, and briefly map the chapters.]

1.3.1 Requirements

1.3.2 Constructing the Solution

Addressing the Requirements

1.3.3 Constraints

Chapter 2

Literature Review

2.1 Review approach

[Describe where and how you searched, search terms, date range, and how you selected sources.]

2.2 Key themes

2.2.1 [Theme 1]

[Compare and synthesize relevant studies; do not just summarize one paper per paragraph.]

2.2.2 [Theme 2]

[Explain agreements, differences, limitations, and what remains unresolved.]

2.3 Research gap and implications for this project

[Show how the reviewed work motivates your objectives and design choices.]

Chapter 3

Progress to Date

3.1 Work completed

[Describe completed investigation, requirements, implementation, or other evidence of progress.]

3.2 Design or investigation alternatives

[Present plausible alternatives and compare them against clear criteria. A table can help.]

Table 3.1: Example comparison of alternatives (replace with your own).

Criterion	Alternative A	Alternative B
[Criterion]	[Evidence / trade-off]	[Evidence / trade-off]

3.3 Preliminary design or investigation

[Describe the current preferred direction, why it was selected, assumptions, and open questions.]

3.4 Challenges and risks

[Explain issues encountered and what you did or will do about them.]

Chapter 4

Research Plan for Next Semester

4.1 Planned work and methods

[List the remaining activities, methods, tools, and expected outputs.]

4.2 Milestones

Table 4.1: Provisional next-semester plan. Replace weeks with actual dates.

Period	Activity	Output
Weeks 1–2	[Refine design / protocol]	[Approved plan]
Weeks 3–5	[Build / collect / investigate]	[Intermediate result]
Weeks 6–8	[Evaluate / analyze]	[Evaluation findings]
Weeks 9–12	[Refine and write up]	[Final report / demonstration]

4.3 Evaluation and success criteria

[State how you will judge whether objectives have been met and what evidence you will collect.]

Chapter 5

Conclusion

[Briefly recap the problem, progress, current direction, and immediate next steps.]

References

[1] H. Asif and J. Vaidya, "A study of users' privacy preferences for data sharing on symptoms-tracking/health app," in *Proceedings of the [conference name]*, Nov. 2022, pp. 109–113. DOI: 10.1145/3559613.3563202

[2] *TipUp: Managing your privacy settings*, en.

[3] Y. Cao, W. Zhang, and C. Han, "How to enhance user privacy via transparent cookie statements," in *Proceedings of the 2025 International Conference on Machine Learning and Neural Networks*, Chongqing China: ACM, Apr. 2025, pp. 336–342.

[4] https://usercentrics.com/knowledge-hub/guide-what-is-a-cmp/, Accessed: 2026-10-30.

[5] Safna, *Cookie consent psychology: What makes users say yes or no?* en, https://www.cookieyes.com/blog/cookie-consent-psychology/, Accessed: 2026-10-30, Jun. 2025.